

STANFORD UNIVERSITY
Department of Electrical Engineering

Lab 1: Output amplifier

Due: Monday, 20 April 2026

Reading: Notes on op-amps and function generators.

Assignment: Design, build, and demonstrate an amplifier that can be used as an output stage for a waveform generator. The amplifier should meet the following specifications:

Output resistance: 50Ω , $\pm 5\Omega$

Input resistance: $>1k\Omega$ (with output terminated in 50Ω)

Unloaded voltage gain magnitude: Adjustable from ~ 0 to 10 (where “ ~ 0 ” means $\ll 1$)

Output-referred DC offset: $< 100mV$

Small-signal bandwidth: DC to $>20MHz$ (with output terminated)

Output overshoot when driven with a $100mV$ input step: $< 5\%$ target, we’ll accept 10%

Output slew rate: $> 100V/\mu s$

Sinusoidal output distortion: $< 3\%$ up to $2MHz$, and up to $5V_{pp}$ across 50Ω load (measured with spectrum analyzer; only need to consider harmonics up to the 5th)

Ability to drive a 50Ω load to at least $\pm 3.2V$ ($20dBm$ sinewave; as a reference point, the Wavetek 166 function generator drives $\pm 7.5V$)

Ability to withstand output shorts to ground indefinitely (the particularly ambitious among you are invited to protect against shorts to both supplies as well). You should show electro-thermal calculations to support your experimental data.

Supply voltages: $\pm 12V$ (we will standardize on this pair of voltages to moderate the number of supplies needed)

Note that the power and bandwidth imply the ability to transmit illegally strong signals in the AM radio band.

Allowed active components: Operational amplifiers (LM301A, uA741, LF356, or LF357), and discrete transistors (2N3904, 2N3906, 2N2219, 2N2905 for BJTs, and J2xx, J3xx and PN4416 for JFETs). These un-exotic, general-purpose commodity devices are chosen intentionally so that your design abilities, rather than device capabilities, are the main determinant of performance. However, if you make a compelling case that remains within the spirit of the design problem, we will certainly consider additional components. You may need to do the “leg work” of procuring components that are not normally stocked here, as Aria Abad isn’t available for more than 24 hours a day, despite appearances to the contrary.

Some possible sources of inspiration to get you thinking:

Take a look at the output amplifiers used in the BK3011B or HP3311A, as they are spec'd similarly to the goals of this design exercise. Don't just copy and paste, **come up with your own design -- this is not a kit-building class**. You should understand (and be able to explain **quantitatively** to the TA) how you approached your design.
